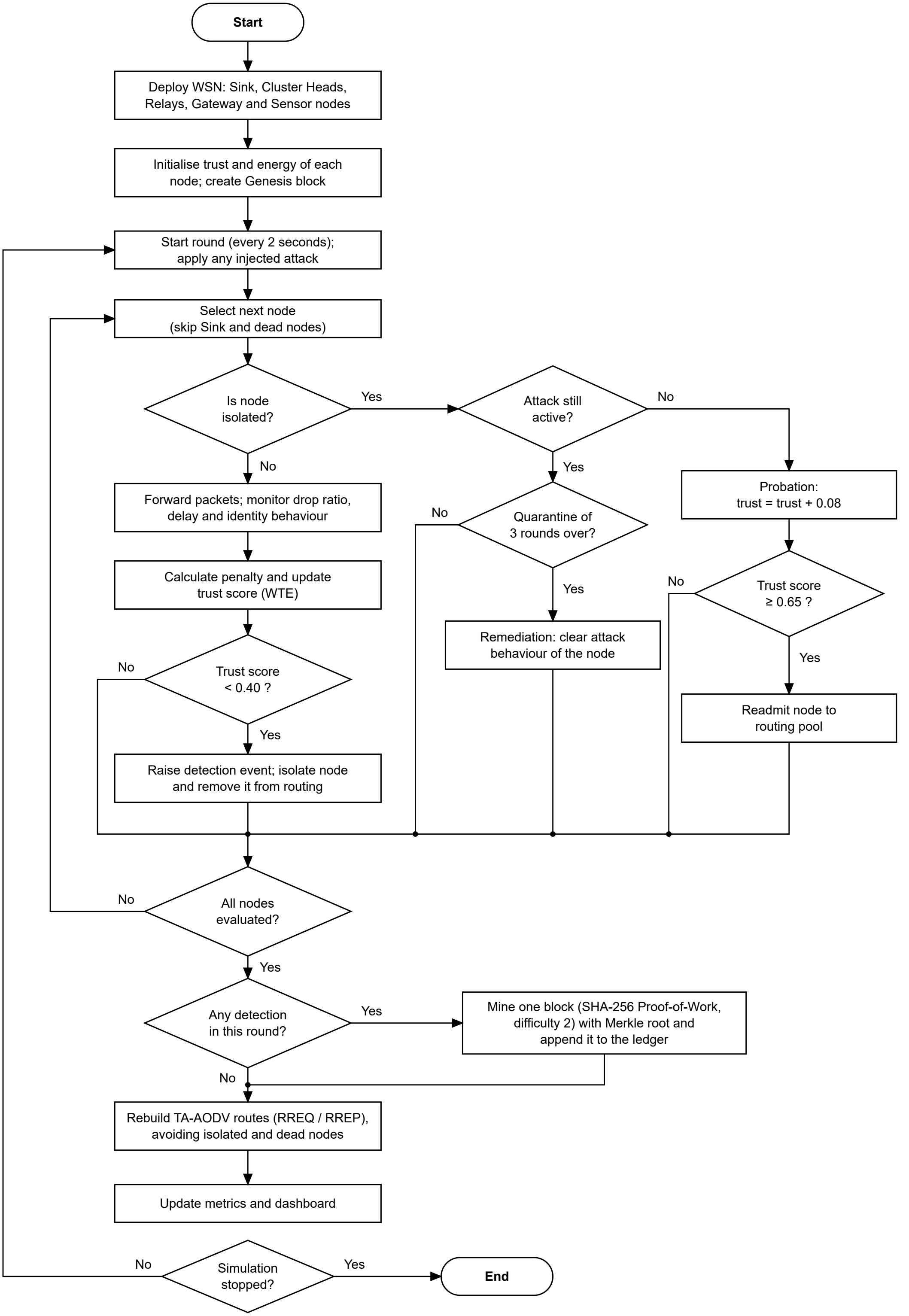

Start
Deploy WSN: Sink, Cluster Heads, Relays, Gateway and Sensor nodes
Initialise trust and energy of each node; create Genesis block
Start round (every 2 seconds); apply any injected attack
Select next node (skip Sink and dead nodes)
Is node isolated?
Yes
No
Attack still active?
No
Yes
Quarantine of 3 rounds over?
No
Yes
Remediation: clear attack behaviour of the node
Probation: trust = trust + 0.08
Trust score ≥ 0.65 ?
No
Yes
Readmit node to routing pool
Forward packets; monitor drop ratio, delay and identity behaviour
Calculate penalty and update trust score (WTE)
Trust score < 0.40 ?
No
Yes
Raise detection event; isolate node and remove it from routing
All nodes evaluated?
No
Yes
Any detection in this round?
Yes
No
Mine one block (SHA-256 Proof-of-Work, difficulty 2) with Merkle root and append it to the ledger
Rebuild TA-AODV routes (RREQ / RREP), avoiding isolated and dead nodes
Update metrics and dashboard
Simulation stopped?
No
Yes
End